SmartInterview: Personalized Technical Interview Preparation in Higher Education

Grounding via Hybrid Graph RAG & Deterministic Bloom's Scaffolding

Slide 1: Title Slide

Project Name: SmartInterview — Personalized Technical Interview Preparation in Higher Education

Project Team ID: Team G-1392

Team Members:

- Shamakura Saiteja Goud (24BD1A665K)
- Thodsam Srujan (24BD1A665R)
- Vasireddy Vignesh Reddy (24BD1A665X)
- Aelugu Ranith Kumar (25BD5A6615)
- Konduri Abhiram (25BD5A6620)

Faculty Guide & Mentor: Dr. TVG Sridevi, Professor

Department: Computer Science and Engineering (AI & ML)

Institution: Keshav Memorial Institute of Technology (KMIT), Hyderabad

Affiliation: Jawaharlal Nehru Technological University Hyderabad (JNTUH)

Category: Edu-Tech / Applied Artificial Intelligence

Slide 2: Literature Gaps & Problem Statement

The Challenge in Modern Technical Recruitment:

- Engineering students face a severe disconnect between theoretical academic syllabi and fast-evolving industry Job Descriptions (JDs).
- High interview anxiety, lack of verbal articulation practice, and high campus recruitment attrition.
- Static coding portals (LeetCode, HackerRank) test binary syntax but cannot simulate architectural trade-off inquiries or verbal dialogues.

Critical Gaps in Foundational 2026 Literature:

- **Base Paper 1 (Nagarajan et al., Feb 2026 — IJMRR):**

* Built strictly for enterprise recruiters as an exclusionary filter.

* Lacks candidate feedback, diagnostic reports, code sandboxes, or curriculum grounding.

- **Base Paper 2 (Wahid et al., March 2026 — IJERT):**

* Silent AI: voice is strictly one-way (Whisper STT only, candidate must read text questions).

* Rigid domain dropdowns without custom JD matching.

- **The Core Gap:** Students lack a grounded, low-latency, two-way conversational coach with deterministic cognitive progression and objective multi-signal grading.

Slide 3: Proposed Solution — SmartInterview

An Adaptive, Grounded Mock Interview & Remediation Platform for Higher Education:

- **3 Distinct Operational Modes:**

1. *Job-Specific Mode:* Set-theoretic Resume-to-JD mapping (Group A Overlap vs. Group B Gaps).
2. *Topic-Based Mode:* Granular technical drilling across 8 core Computer Science domains.
3. *Semester Syllabus Mode:* Isolated vector collections generated dynamically from university syllabus PDFs.

- **Hybrid Graph RAG Grounding:** Combines Dense Semantic ChromaDB, BM25Okapi Lexical Search, and NetworkX Knowledge Graph Traversal to completely eliminate technical hallucinations.
- **Deterministic Cognitive Scaffolding:** State-driven difficulty scaling via Bloom's Revised Taxonomy (*Remember* through *Create*).
- **Objective Multi-Signal Evaluation:** 5-factor mathematical rubric eliminating subjective LLM "vibe grading".
- **Remedial Mastery Mode:** Automated weakness diagnosis, prerequisite ontology traversal, and 1-click focused retests.

Slide 4: Decoupled 4-Tier Distributed System Architecture

Visual Asset: `SmartInterview_Deliverables/05_System_Diagrams/Architecture.png`

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Key Points:

- Enforces strict boundary between local zero-cost persistence and accelerated cloud inference.
- Operates smoothly on commodity laboratory hardware without requiring local GPUs.

Slide 5: Core Workflow & Context Ingestion Pipeline

Visual Asset: `SmartInterview_Deliverables/05_System_Diagrams/Workflow.png`

Personalized Technical Staging Pipeline:

1. **Document Ingestion:** Candidate uploads Resume (PDF) and target Job Description (or Syllabus).
2. **Deterministic Extraction:** PyMuPDF isolates technical competencies, frameworks, and project details.
3. **Set-Theoretic Skill Gap Analysis:**
   - *Group A (Role Overlap):* Skills present in both Resume & JD -> Targeted for deep technical verification.
   - *Group B (Role Gaps):* Skills required by JD but missing in Resume -> Targeted for adaptability & breadth.
4. **Targeted Question Staging:** Generates contextual questions grounded in verified textbook reference chunks.
5. **Real-Time Dialogue & Multi-Signal Evaluation:** Transcribes candidate audio and scores performance across 5 mathematical signals.
6. **Bloom Cognitive Escalation / Remediation:** Advances or regresses cognitive levels dynamically based on mastery.

Slide 6: Hybrid Graph RAG Architecture & Ontology

Tri-Retrieval Engine with Reciprocal Rank Fusion (RRF):

- **The Challenge:** Vanilla vector search misses exact technical keywords (**`2PL`**, **`ACID`**) and cannot traverse conceptual prerequisites.
- **Tri-Retrieval Pipeline:**

1. *Dense Semantic Retrieval (ChromaDB + all-MiniLM-L6-v2):* 384-dimensional cosine index across 402 curated CS topic chunks in 8 domains.

2. *Lexical Keyword Retrieval (BM25Okapi):* Deterministic exact token matching for symbols, algorithms, and acronyms.

3. *Knowledge Graph Traversal (NetworkX In-Memory / Neo4j):* Directed graph mapping 50 CS concepts, prerequisite edges, and Bloom cognitive depths (**`data/graph_data.json`**).

- **Reciprocal Rank Fusion (RRF, $k=60$):**

$$RRF(d) = \sum_{m \in M} \frac{1}{60 + r_m(d)}$$

- **Empirical Benchmark Results (33 Technical Domain Queries):**
- **Hit@1 = 96.97%** (Top retrieved chunk is the exact ground-truth answer key).
- **Hit@3 = 100.00%** (Correct reference chunk guaranteed in top 3).
- **Mean Reciprocal Rank (MRR) = 0.9798** (Near-perfect ranking precision).
- **Retrieval Latency = 24.6 ms** (Sub-100ms vector search via local ChromaDB HNSW).

Slide 7: Relational Data Schema & Entity Architecture (ERD)

Visual Asset: `SmartInterview_Deliverables/05_System_Diagrams/ERD.png`

Robust 7-Table Relational Schema (MySQL 8.0 InnoDB):

- **`users`**: Candidate credentials, bcrypt password hash, timestamps.
- **`resumes`**: PyMuPDF parsed skills, work experience, projects JSON.
- **`job_descriptions`**: Target JD entities, domain skills, requirements JSON.
- **`interview_sessions`**: Session difficulty, question count, status, serialized **`AdaptiveState`** JSON, Bloom level.
- **`interview_questions`**: Question text, Bloom tier, skill, reference RAG chunk ID.
- **`answers`**: Candidate audio transcription and response text.
- **`answer_evaluations`**: Multi-signal 5-factor scores, feedback, strengths, weaknesses.
- **Guarantees:** Foreign keys with **`ON DELETE CASCADE`**, strict ACID transactions, and JSON state serialization preventing session loss.

Slide 8: Deterministic Cognitive Scaffolding (Bloom's Taxonomy)

Strict Cognitive Progression (Replacing Arbitrary LLM Difficulty):

- Maps questions systematically across **Bloom’s Revised Taxonomy**:

Remember (L1) -> Understand (L2) -> Apply (L3) -> Analyze (L4) -> Evaluate (L5) -> Create (L6)*.

- **Deterministic State Transition Logic:**
- **Score $\ge$ 80% (Mastery):** Advance to higher Bloom level (+1) or advance difficulty tier (*Easy -> Medium -> Hard*).
- **Score 50% – 79% (Adequate):** Maintain current Bloom level and difficulty tier for concept reinforcement.
- **Score < 50% (Remediation):** Regress Bloom level (-1) or lower difficulty to rebuild core principles.
- **Boundary Protections:**
- *Ceiling Protection:* At Create (L6) + Hard, scores maintain maximum cognitive depth.
- *Floor Protection:* At Remember (L1) + Easy, floors cleanly without negative underflow.
- **Weakness-Biased Skill Selection Algorithm:** Automatically samples next questions from lowest-scoring skills to eliminate competency blind spots.

Slide 9: Multi-Signal Answer Evaluation (Eliminating "Vibe Grading")

Objective, Multi-Dimensional Scoring Rubric:

- Evaluates candidate answers using a strict 5-part weighted mathematical formula:

* **30% Technical Correctness:** Deep LLM logical reasoning and factual accuracy verified against retrieved ground truth.

* **20% Completeness:** Coverage of all question constraints, trade-offs, and edge cases.

* **20% Relevance:** Focus, conciseness, and absence of conversational filler rhetoric.

* **15% Semantic Cosine Similarity:** Sentence-BERT (`all-MiniLM-L6-v2`) matching candidate answer against verified ground-truth chunk.

* **15% Explicit Concept Coverage:** Deterministic keyword and entity regex detection for mandatory technical terms.

- **Why This Split?** Blends semantic understanding with hard concept verification, ensuring reproducible, unbiased scoring immune to prompt drift.

Slide 10: Live Spoken Interview Room & Neural Voice

Visual Asset: `screenshots/13_live_interview_session.png`

Real-Time Two-Way Conversational Dialogue:

- **Text-to-Speech (Microsoft Edge-TTS):**

* Professional **`en-US-ChristopherNeural`** technical recruiter voice profile.

* Synthesized directly into in-memory **`io.BytesIO`** buffers (latency $\le 1.2\text{s}$, zero disk footprint).

- **Speech-to-Text (Groq Whisper Large-v3):**

* High-accuracy neural speech transcription conditioned on CS domain vocabulary priming.

- **Interactive Visual Interface:**

* Displays active Bloom taxonomy level badge (e.g. **`Analyze`** Level 4).

* Real-time speech recognition status, waveform audio controls, and fallback text area.

* Sub-2.5s end-to-end turnaround eliminates awkward conversational pauses.

Slide 11: Candidate Evaluation Scorecard & Analytics

Visual Asset: `screenshots/11_results_scorecard.png`

Multi-Dimensional Candidate Feedback:

- **Overall Session Score & Performance Tier:** Percentage score with Bloom cognitive depth indicators.
- **5-Axis Radar Breakdown Chart:** Visualizes Correctness, Completeness, Relevance, Semantic Similarity, and Concept Coverage.
- **Granular Question Transcripts:** Side-by-side comparison of candidate answers with verified reference context.
- **Strengths & Improvement Areas:** Concrete itemized feedback highlighting exact technical gaps.
- **Actionable Recommendations:** AI-curated study priorities tagged by urgency.
- **Downloadable Formal PDF Report:** 1-click export of comprehensive academic interview report.

Slide 12: Remedial Mastery Mode & Targeted Upskilling

Visual Asset: `screenshots/12_remedial_mastery_mode.png`

Transforming Assessment into Active Learning:

- **Knowledge Graph Root-Cause Tracing:**

* When candidate scores < 50%, engine queries NetworkX graph for missing prerequisite concepts.

- **Automated AI Curriculum & Study Guide Synthesis:**

* Generates high-yield study notes, core concepts, and "where candidates typically lose marks".

- **Isolated Prerequisite Knowledge Chunks:**

* Retrieves exact textbook references for targeted review.

- **1-Click Focused Retest Workflow:**

* Candidates launch an immediate retest on failed concepts to achieve mastery.

Slide 13: System Verification: 355-Test Matrix (100% Passing)

Exhaustive Test Coverage Across All 18 Engineering Phases:

- **355 Total Automated Unit, Integration & Validation Tests (100% Pass Rate in 87.90s):**

Phase 1:* Dense Vector RAG Baseline & S-BERT embeddings (25 tests)

Phase 2-4:* Knowledge Graph Schema, Builder, Storage & Canonical Chunk Resolution (69 tests)

Phase 5:* BM25 Lexical Keyword Retrieval Engine & Tokenization (34 tests)

Phase 6-7:* Graph Traversal Retrieval & Hybrid Reciprocal Rank Fusion (48 tests)

Phase 8:* Reranker Validation & Multi-Signal Scoring (36 tests)

Phase 9-10:* Pipeline Integration & Bloom's Cognitive Progression (39 tests)

Phase 11-13:* Security Audit, Evaluation Metrics & Golden Dataset Auditing (48 tests)

Phase 14-18:* Frozen Baseline, Hybrid Evaluation & Edge Case Safeguards (56 tests)

- **Key Verified Functional Gates:**

* **`TC-01`**: Duplicate auth rejected (HTTP 400)

* **`TC-02`**: Corrupt uploads safely handled

* **`TC-03`**: Sub-25ms retrieval latency

* **`TC-04`**: Deterministic Bloom state escalation ($\ge 80\%$) & regression ($< 50\%$)

* **`TC-05`**: Remedial graph prerequisite traversal

* **`TC-06`**: In-memory Edge-TTS $\le 1.2\text{s}$

* **`TC-07`**: 5-signal rubric convergence

* **`TC-08`**: Starvation safeguards & fallback handling

Slide 14: Conclusion & Viva Q&A

Conclusion:

SmartInterview bridges the gap between theoretical higher education and dynamic industry hiring. By combining Hybrid Graph RAG grounding (Hit@1: 96.97%, Hit@3: 100%), deterministic Bloom's taxonomy cognitive scaffolding, Remedial Mastery drills, and multi-signal objective evaluation, we deliver an authentic, unbiased, low-latency interview preparation ecosystem at zero infrastructure cost.

Thank You! We are now open for questions and technical discussion from the honorable panel.